Highlights

Detection of Anxiety, Depression, and Eating Disorders Using Transformer-Based Language Models: A Comparative Study from BERT Encoders to Frontier LLMs

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- Twelve language models are systematically compared across four mental health classification tasks, integrating performance, stability, statistical significance, and cost.
- Large language models achieve the highest performance across all tasks, with macro F1-scores reaching between 0.842 and 0.979.
- BERT-based encoders remain highly competitive, with only small performance gaps compared with the best LLMs in most classification tasks.
- Most performance differences among the strongest models are not statistically significant, despite differences observed in their point estimates.
- Cost-performance analysis shows that larger and more expensive language models do not necessarily provide better practical value.

Detection of Anxiety, Depression, and Eating Disorders Using Transformer-Based Language Models: A Comparative Study from BERT Encoders to Frontier LLMs

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Abstract

This study compares fine-tuned BERT-based encoders, open-source large language models (LLMs), and proprietary frontier LLMs for detecting anxiety, depression, and eating disorders. Twelve models were evaluated on the MentalRiskES benchmark, comprising 500 users for anxiety detection, 499 for depression, and 335 for eating disorders, using three binary tasks and a multiclass setting. LLMs achieved the highest macro F1-score in all tasks, reaching 0.900 for anxiety, 0.842 for depression, 0.979 for eating disorders, and 0.849 for multiclass classification. The largest advantage over the best BERT-based encoder was observed for anxiety (0.900 vs. 0.680), whereas the gaps for the remaining tasks did not exceed 0.04. Paired statistical comparisons showed that most differences among the strongest models were not statistically significant, while all evaluated LLMs exhibited high prediction stability, with TARa@5 values above 0.83. Error analysis identified Control, Anxiety, and Depression as the main sources of confusion, whereas eating disorders were the most readily distinguishable condition. Cost-performance analysis showed that RoBERTa-large-BNE and Gemini 3 Flash offered particularly favorable trade-offs. Overall, the results show that recent LLMs can

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improve mental health disorder detection, but well-performing BERT-based encoders remain competitive when statistical uncertainty and computational cost are considered.

Keywords: Mental health, BERT, Large language models, Text classification, Natural language processing

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1. Introduction

Mental health disorders represent one of the most pressing public health challenges worldwide (World Health Organization, 2022). According to the World Health Organization, more than one billion people are affected by a mental disorder, with anxiety, depression, and eating disorders (EDs) among the most prevalent conditions. These disorders impose a substantial personal, social, and economic burden, making early identification a key factor for improving clinical outcomes (Cuijpers et al., 2023). However, conventional assessment methods rely heavily on trained mental health professionals and are often time-consuming and resource-intensive, limiting their scalability (World Health Organization, 2022). Meanwhile, the widespread adoption of social media and online communication platforms has produced vast amounts of user-generated text containing valuable linguistic and behavioral signals. This has motivated the development of Natural Language Processing (NLP) techniques for the automatic identification of mental health indicators from textual data (Ge et al., 2025).

Recent advances in transformer-based language models have significantly expanded the capabilities of NLP systems (Zhao et al., 2026). Following the introduction of the Transformer architecture, encoder-based models such as BERT and its variants achieved state-of-the-art performance in text classification through task-specific fine-tuning (Devlin et al., 2019; Liu et al., 2019). More recently, the emergence of Large Language Models (LLMs) has introduced a new paradigm, enabling models to perform downstream tasks through zero-shot or few-shot prompting without requiring supervised retraining (Brown et al., 2020; Chae and Davidson, 2026). Owing to their strong language understanding and reasoning capabilities, these models have attracted increasing interest for mental health applications, where subtle linguistic patterns can provide important clues about an individual's psychological state.

Several studies have demonstrated the effectiveness of transformer-based

models for the automatic detection of mental health disorders. Yang et al. (2023) showed that domain-specific transformer models such as MentalBERT and MentalRoBERTa consistently outperform recurrent architectures, with MentalRoBERTa achieving F1-scores of 94.23% for depression detection, 81.76% for stress detection, and 89.01% in multiclass mental health classification. Similarly, Hasan et al. (2025) compared recurrent neural networks and transformer models on a large Reddit dataset, reporting that RoBERTa achieved 91–99% F1-score and accuracy depending on the diagnostic category, confirming the superior generalization capabilities of transformer-based approaches. Yang et al. (2025) further demonstrated that combining transformer models with data augmentation, class balancing, and feature fusion techniques substantially improves multiclass classification performance, increasing RoBERTa's F1-score from 0.816 to 0.857 and BERT's from 0.766 to 0.810.

Research focusing on Spanish remains considerably more limited. The introduction of the MentalRiskES dataset (Mármol-Romero et al., 2024) has provided one of the first benchmarks for Spanish mental health disorder detection. Together with the dataset, Mármol-Romero et al. reported baseline results using transformer-based encoders such as RoBERTa and mDeBERTa, establishing an initial benchmark for future research. Subsequently, Pan et al. (2023) evaluated additional monolingual transformer models on the MentalRiskES benchmark, showing that language-specific models such as BETO and MarIA outperform multilingual alternatives, achieving macro F1-scores of 0.737 for depression detection and 0.918 for eating disorder detection.

The rapid development of LLMs has also motivated numerous studies exploring their potential in mental health assessment. However, current evidence remains mixed. Kallstenius et al. (2025) compared traditional NLP approaches, zero-shot prompting with GPT-4o-mini, and a fine-tuned version of the same model on a multiclass mental health dataset comprising more than 50,000 posts. Surprisingly, a conventional TF-IDF + SVM classifier achieved 95% accuracy, outperforming both zero-shot GPT-4o-mini (65%) and its fine-tuned counterpart (91%). In contrast, Liu et al. (2025) proposed EmoScan, a Mistral-7B model fine-tuned specifically for anxiety and depression screening, achieving an F1-score of 0.75 on the primary classification task and 0.67 on an external evaluation dataset. Likewise, Danner et al. (2023) evaluated both BERT and GPT-3.5 for depression detection from clinical interviews, reporting F1-scores of 0.82 and 0.78, respectively, highlighting the competitiveness of LLMs while confirming the advantage of

Table 1: Comparison of representative studies on automatic mental health disorder detection using NLP and LLMs.

Study	Models	Dataset / Language	Classification Tasks	Best Reported F1-score
Yang et al. (2023)	MentalBERT, MentalRoBERTa	Reddit / English	Depression, Stress, Multiclass	0.942, 0.818, 0.890 (MentalRoBERTa)
Yang et al. (2025)	LSTM, BERT, RoBERTa	CounselChat + augmentation / English	Multiclass	0.857 (RoBERTa)
Mármol-Romero et al. (2024)	RoBERTa Base, RoBERTa Large, mDeBERTa	MentalRiskES / Spanish	Anxiety, Depression, EDs	0.710 (mDeBERTa), 0.700 (RoBERTa Large), 0.810 (mDeBERTa)
Pan et al. (2023)	BETO, MarIA	MentalRiskES / Spanish	Depression, EDs	0.737 (BETO), 0.918 (MarIA)
Liu et al. (2025)	EmoScan (Mistral-7B)	PsyInterview / English	Anxiety, Depression	0.750 (EmoScan)
Danner et al. (2023)	BERT, GPT-3.5	DAIC-WOZ + Extended-DAIC / English	Depression	0.820 (BERT), 0.780 (GPT-3.5)

task-specific encoder models.

More recent studies have explored broader applications of LLMs beyond conventional text classification. Xu et al. (2025) combined fine-tuned LLMs with linguistic and acoustic features to analyse psychiatric interviews from 1,160 outpatients, achieving 86.9% accuracy for clinical annotation identification and 74.7% and 77.2% accuracy for anxiety and depression detection, respectively. In parallel, Yang et al. (2024) introduced MentalLLaMA, a family of LLaMA2-based models specifically adapted to mental health applications. Trained on more than 105,000 instruction-tuning examples, MentalLLaMA achieved F1-scores above 0.80 across multiple classification tasks while producing explanations comparable to GPT models, with performance remaining within 5% of state-of-the-art discriminative approaches.

Table 1 summarizes representative studies on automatic mental health disorder detection, highlighting the evaluated models, datasets, classification tasks, and best reported performance.

Despite these advances, several important limitations remain. First, most existing studies focus on English-language datasets, with comparatively little research conducted on Spanish corpora. Second, depression has received substantially more attention than anxiety and eating disorders, which are often underrepresented or omitted entirely. In addition, many publicly available datasets rely on self-reported diagnoses or participation in online communities, potentially introducing annotation noise and bias.

From a methodological perspective, several challenges also remain. Although a growing number of studies have compared transformer-based models with LLMs, these evaluations have typically focused on a limited number of models or early generations of LLMs. Comprehensive comparisons spanning traditional BERT-based encoders together with the latest generation of proprietary frontier and open-source LLMs remain scarce. Furthermore, relatively few studies assess predictive stability, statistical significance, and the trade-off between predictive performance and inference cost within a unified evaluation framework.

1.1. Research Objectives and Questions

To address these gaps, this study presents a comprehensive evaluation of transformer-based language models for automatic mental health disorder detection in Spanish. The study compares representative BERT-based encoders with the latest generation of proprietary frontier and open-source LLMs across anxiety, depression, and eating disorder detection tasks. Beyond conventional performance metrics, the evaluation examines prediction stability, systematic error patterns, the statistical significance of performance differences, and the trade-off between predictive performance and inference cost. Together, these analyses provide a comprehensive assessment of the strengths, limitations, and practical applicability of current language models for mental health disorder detection.

To guide this evaluation, the study addresses four research questions: (1) How do traditional BERT-based encoder models, open-source LLMs, and proprietary frontier LLMs compare in the automatic detection of anxiety, depression, and eating disorders in Spanish? (2) What systematic error patterns emerge across different model families, and which mental health conditions are most frequently confused in multiclass classification? (3) Are the observed performance differences between models statistically significant, and how stable are LLM predictions across repeated executions? and (4) What is the relationship between inference cost and predictive performance, and which models provide the most favorable cost-performance trade-off?

2. Methods

2.1. Dataset

The experiments were conducted using the MentalRiskES dataset (Mármol-Romero et al., 2024), a Spanish corpus specifically developed for automatic

Table 2: Class distribution of the MentalRiskES dataset.

Task	Control	Positive
Anxiety	57	443
Depression	166	333
Eating disorders	192	143

mental health disorder detection. The dataset comprises anonymized Telegram users whose chronological messages were collected from public groups related to anxiety, depression, and eating disorders (EDs). Each user constitutes the unit of analysis and is represented by the complete sequence of messages posted within a given discussion group, allowing models to exploit the broader linguistic context rather than isolated messages.

A key characteristic of MentalRiskES is its annotation methodology. Instead of relying solely on self-reported diagnoses or community membership, each user was independently annotated by ten native Spanish speakers following standardized annotation guidelines based on official clinical definitions. Final labels were assigned by majority voting. Although the annotators were not mental health professionals, the use of multiple independent annotators and a standardized annotation protocol provides greater reliability than many publicly available mental health datasets. Nevertheless, the resulting labels should be interpreted as indicators of mental health risk rather than clinical diagnoses.

Table 2 summarizes the class distribution for the three binary classification tasks. The dataset contains 500 users for anxiety detection, 499 for depression detection, and 335 for eating disorder detection. Class imbalance varies considerably across tasks, with anxiety presenting the most pronounced imbalance (443 positive vs. 57 control users), whereas eating disorders exhibit a more balanced distribution. The multiclass dataset was constructed by combining the samples from the three binary datasets into a single four-class classification task comprising Control, Anxiety, Depression, and Eating Disorders.

Although MentalRiskES is one of the few Spanish mental health datasets with a carefully designed multi-annotator labeling process, its limited size poses an important challenge. The small number of annotated users may hinder the fine-tuning of supervised models and reduce the statistical power of performance comparisons. This limitation further motivates the inclu-

sion of statistical significance testing and predictive stability analyses in the experimental evaluation.

2.1.1. Embedding visualization

Before training and evaluating the models, an exploratory analysis was conducted to assess the intrinsic separability of the multiclass dataset. User-level texts were encoded using the multilingual *paraphrase-multilingual-mpnet-base-v2* sentence embedding model (Reimers and Gurevych, 2020) and projected into a two-dimensional space using t-distributed Stochastic Neighbor Embedding (t-SNE) (Maaten and Hinton, 2008).

Figure 1 shows that the different mental health conditions exhibit varying degrees of semantic overlap. Eating disorder samples form a relatively compact and well-separated cluster, whereas anxiety and depression show substantial overlap, suggesting that these conditions share similar linguistic characteristics. The control class is distributed across several regions of the embedding space, reflecting its greater linguistic heterogeneity. These observations suggest that the eating disorder class is more easily separable than the remaining classes and provide qualitative evidence of the inherent difficulty of distinguishing anxiety from depression using textual information alone.

2.2. Evaluated models

To provide a comprehensive comparison across different language model paradigms, the evaluation includes three complementary families of transformer-based models: (1) encoder-based models derived from BERT, (2) open-source LLMs, and (3) proprietary frontier LLMs. Rather than exhaustively evaluating all available models, the selection aims to cover representative architectures with different characteristics in terms of specialization, scale, and deployment requirements.

The encoder-based models include DistilBETO (Cañete et al., 2022), RoBERTa-large-BNE (Gutiérrez-Fandiño et al., 2022), and mDeBERTa-v3-base (He et al., 2021). DistilBETO and RoBERTa-large-BNE were selected as representative Spanish-specific encoders, covering both lightweight and large-scale architectures, while mDeBERTa-v3-base provides a strong multilingual baseline. This selection makes it possible to assess the impact of language specialization and model capacity within the encoder paradigm.

The open-source LLMs comprise Llama 3.1 8B (Meta-AI, 2024), GPT-OSS 20B (Agarwal et al., 2025), and DeepSeek-V3.2 (DeepSeek-AI, 2025),

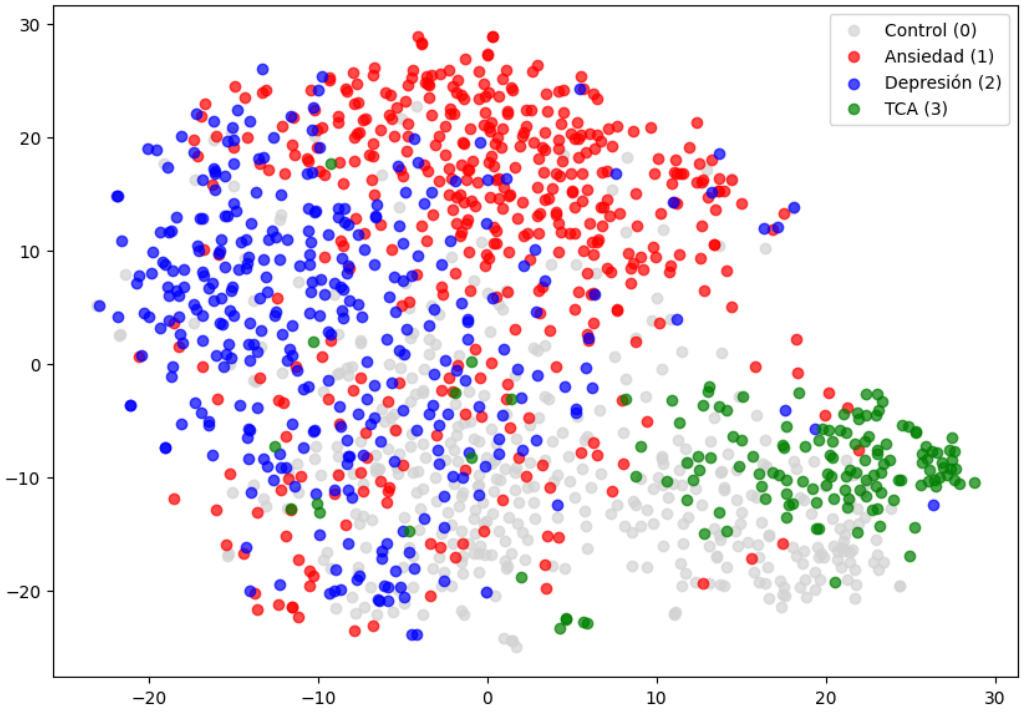


Figure 1: Two-dimensional visualization of multilingual sentence embeddings for the multiclass MentalRiskES dataset using t-SNE.

representing models with different parameter scales and training strategies. Together, they provide a representative sample of the current open-source ecosystem, ranging from relatively compact models suitable for local deployment to larger models optimized for higher reasoning performance.

Finally, the proprietary models include the latest frontier and lightweight variants available at the time of the study from the three leading commercial AI providers: OpenAI (GPT-5.2 and GPT-5 Mini), Anthropic (Claude Sonnet 4.6 and Claude Haiku 4.5), and Google (Gemini 3.1 Pro Preview and Gemini 3 Flash Preview). Including both flagship and lightweight variants enables the evaluation of the trade-offs between predictive performance, computational cost, and inference efficiency within each model family.

Table 3: Model-specific batch sizes used for encoder fine-tuning.

Model	Train Batch	Eval Batch
DistilBETO	16	64
RoBERTa-large-BNE	8	32
mDeBERTa-v3-base	8	32

2.3. Experimental Setup

2.3.1. BERT-Based Encoder Fine-Tuning

The encoder-based models were fine-tuned using the Hugging Face Transformers library (Wolf et al., 2020). Training was performed on a NVIDIA T4 GPU provided by Google Colab, which imposed memory constraints for the largest encoder architectures. For each classification task, the dataset was divided into training (70%), validation (15%), and test (15%) subsets using stratified sampling to preserve the original class distribution. The validation set was used for model selection, while the test set remained unseen until the final evaluation.

User-level texts were tokenized using the tokenizer associated with each pretrained model and truncated to a maximum sequence length of 512 tokens. The classification head was adapted to the target task, using two output neurons for the binary experiments and four output neurons for the multiclass setting.

To ensure a fair comparison, all encoder models were trained using the same optimization strategy and hyperparameter configuration. Fine-tuning was performed for a maximum of 10 epochs using the AdamW optimizer with a learning rate of 2×10^{-5}, a warm-up ratio of 0.1, and a weight decay of 0.01. To mitigate class imbalance, class weights computed from the training set were incorporated into a weighted cross-entropy loss function. Model selection was based on the macro F1-score obtained on the validation set, and early stopping with a patience of two epochs was applied to reduce overfitting. The best-performing checkpoint was subsequently evaluated on the test set.

The only model-specific hyperparameter was the batch size, which was adjusted according to the memory requirements of each architecture while keeping all remaining training settings unchanged. Table 3 summarizes the batch sizes used during training and evaluation.

2.3.2. Zero-Shot Prompting

The LLMs were evaluated using a zero-shot prompting strategy, without task-specific fine-tuning or parameter updates. Instead, each model received a natural language instruction describing the classification task together with the complete set of messages associated with a user, and was required to predict the corresponding class directly from its pretrained knowledge.

To ensure a fair comparison with the encoder-based models, all LLMs were evaluated on exactly the same test partitions used during BERT fine-tuning. Consequently, the LLMs generated predictions only for the held-out 15% test set of each classification task, while the remaining training and validation subsets were not used during inference. This protocol guarantees that all reported performance differences are based on identical evaluation samples.

To ensure a fair comparison across LLMs, a common inference protocol was adopted. The same prompt structure was used for all models, with task-specific adaptations for the binary and multiclass settings. Each prompt included a brief description of the target disorder(s), the complete user text, and explicit instructions restricting the output to the corresponding numerical class label. All proprietary models were accessed through their official APIs, while the open-source models were evaluated through OpenRouter, which provides a standardized API interface across different providers. The temperature was fixed to 0 for all experiments to maximize determinism and reduce variability due to stochastic generation.

The prompts were designed to provide only the task definition and the input text, without annotated examples or additional contextual information. Consequently, model predictions relied exclusively on the knowledge acquired during pretraining. Using an identical prompting strategy for all evaluated LLMs ensures that performance differences primarily reflect the intrinsic capabilities of the models rather than differences in prompt engineering.

2.4. Evaluation Methodology

The experimental evaluation follows the general framework proposed by Domínguez-Díaz et al. (2026), with several methodological adaptations tailored to the objectives of this study. These include updated statistical comparison procedures and the incorporation of a cost-performance evaluation.

All scripts required to reproduce the experiments, statistical analyses, figures, and tables reported in this paper are publicly available in the accompanying code repository (see the Code availability section).

2.4.1. Classification Metrics and Prediction Stability

Model performance was primarily evaluated using macro F1-score, which assigns equal importance to all classes and is well suited to imbalanced classification problems (Sokolova and Lapalme, 2009). Precision, recall, and the Matthews Correlation Coefficient (MCC) were additionally reported to provide a more comprehensive assessment of classification performance (Matthews, 1975). To quantify the uncertainty associated with the reported metrics, 95% bias-corrected and accelerated (BCa) bootstrap confidence intervals were estimated using 1000 bootstrap resamples (Efron, 1992, 1987).

Since LLMs may exhibit non-deterministic behavior even under identical inference settings, their prediction stability was additionally assessed using the TARa@5 metric (Atil et al., 2025). For each test instance, every LLM generated five independent predictions using the same prompt and inference configuration. TARa@5 was computed as the average agreement between each prediction and the modal prediction across all instances. Owing to the discrete agreement counts underlying this metric, uncertainty was quantified using 95% Bayesian credible intervals estimated with a Beta-Binomial model (Gelman et al., 1995), which provides a more appropriate probabilistic treatment than bootstrap-based confidence intervals for this type of data. Higher TARa@5 values indicate greater prediction stability across repeated executions. As encoder-based models perform deterministic inference after fine-tuning, this analysis was applied exclusively to the evaluated LLMs.

2.4.2. Error Analysis

Overall performance metrics do not reveal which classes are systematically confused by the evaluated models. Therefore, model predictions were further analysed using row-normalized confusion matrices, allowing the identification of recurrent misclassification patterns independently of class frequencies. This analysis is particularly relevant for the multiclass task, where the semantic similarity between anxiety and depression is expected to increase the difficulty of class discrimination. The confusion matrices complement the quantitative metrics by providing a qualitative assessment of model behaviour across different mental health conditions.

2.4.3. Paired Model Comparisons

Statistical comparison of top-performing models. Beyond reporting individual performance metrics, pairwise statistical comparisons were conducted to determine whether the observed differences between models were statistically

significant. To keep the number of comparisons manageable while focusing on the most competitive systems, statistical testing was restricted to a subset of the best-performing models for each classification task. Specifically, the comparison included the highest-ranked BERT-based encoder, the best-performing open-source LLM, and the two best-performing proprietary LLMs according to their macro F1-score.

Pairwise comparisons were performed using the Approximate Randomization Test (ART) (Yeh, 2000), a non-parametric permutation test widely adopted in natural language processing for comparing classifiers evaluated on the same test instances. ART assesses whether the observed difference in macro F1-score between two models is likely to have occurred by chance by repeatedly randomizing their predictions while preserving the ground-truth labels. In this study, 10,000 randomization iterations were used to estimate the corresponding p-values.

In addition to statistical significance, the difference in macro F1-score between each pair of models was quantified together with a 95% bootstrap confidence interval obtained from 5,000 paired bootstrap resamples (Efron and Tibshirani, 1994; Davison and Hinkley, 1997). This interval provides an estimate of both the magnitude and the uncertainty associated with the observed performance difference.

Since multiple pairwise comparisons were performed, p-values obtained from ART were adjusted using the Benjamini-Hochberg procedure to control the false discovery rate (Benjamini and Hochberg, 1995).

McNemar test. For the three binary classification tasks, statistical comparisons were further complemented using the exact McNemar test (McNemar, 1947). Unlike ART, which compares differences in overall predictive performance, McNemar evaluates whether two classifiers exhibit significantly different error patterns by analysing their disagreements on individual test instances. Because the multiclass task involves more than two classes, McNemar was not applied in that setting.

Intra-family comparison of proprietary LLMs. In addition to the comparison between the best-performing models, a second analysis was conducted within each proprietary LLM family to investigate whether the flagship models provide statistically significant improvements over their lightweight counterparts. Three pairwise comparisons were therefore performed: GPT-5.2 vs. GPT-5 Mini, Claude Sonnet 4.6 vs. Claude Haiku 4.5, and Gemini 3.1

Pro vs. Gemini 3 Flash. The same statistical methodology described above (ART, bootstrap confidence intervals, and McNemar for binary tasks) was applied to each pair. This analysis provides a practical assessment of whether the additional computational cost of larger proprietary models translates into meaningful performance improvements.

2.4.4. Cost–Performance Analysis

Beyond predictive performance, the practical efficiency of the evaluated models was assessed by analysing the relationship between classification performance and computational cost.

For the LLMs, inference costs were obtained directly from the official pricing of the corresponding API providers at the time the experiments were conducted. The total cost for each classification task was computed from the actual API usage, taking into account both input and output tokens.

For the BERT-based encoder models, no API usage costs were involved. Instead, the computational cost was estimated from the total execution time required for fine-tuning and inference on each task. Execution times were multiplied by a reference GPU rental cost of USD 0.41 per hour, corresponding to the approximate on-demand price of an NVIDIA T4 GPU on Google Cloud. Although this estimate does not represent the exact monetary cost of every possible deployment environment, it provides a consistent basis for comparing the computational resources required by encoder-based models.

3. Results

This section first summarizes the best-performing BERT-based encoder and LLM across the four classification tasks. The subsequent analyses focus on the multiclass setting, which constitutes the most challenging and clinically relevant scenario. This setting enables a detailed examination of prediction stability, error patterns, statistical differences, and cost-performance trade-offs. Complete results for the binary tasks are available in the accompanying repository.

3.1. Overall Performance Across Classification Tasks

Figure 2 summarizes the highest macro F1-score achieved by the best-performing BERT-based encoder and LLM for each classification task. In the case of the LLMs, ties in macro F1-score were resolved by selecting the

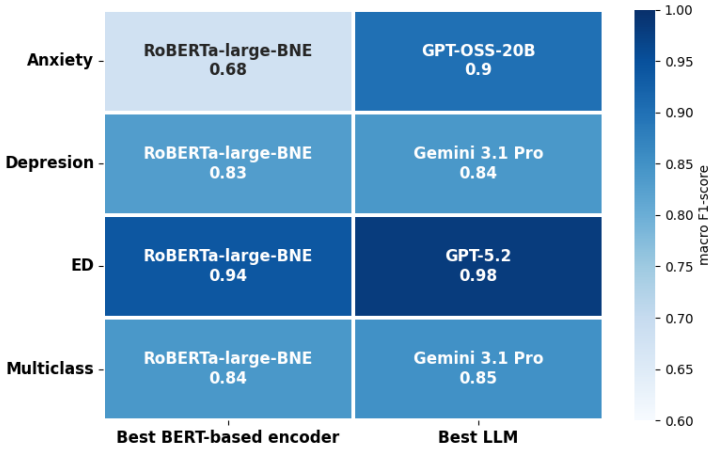


Figure 2: Best macro F1-score obtained by the highest-performing BERT-based encoder and LLM for each classification task.

model with the higher upper bound of the corresponding 95% confidence interval for visualization purposes only.

Overall, LLMs achieved the highest macro F1-score in all four tasks, although the performance advantage over BERT-based encoders varied substantially across disorders. The largest improvement was observed for anxiety detection, where GPT-OSS-20B outperformed the best encoder model by 0.22 macro F1 points (0.90 vs. 0.68). This marked difference may be related to the pronounced class imbalance of the anxiety dataset, which appears to affect the supervised fine-tuning of encoder-based models more strongly. By contrast, the performance gap was considerably smaller for depression (0.84 vs. 0.83), eating disorder detection (0.98 vs. 0.94), and multiclass classification (0.85 vs. 0.84), suggesting that encoder-based models remain highly competitive for these tasks.

Among the evaluated encoder models, RoBERTa consistently achieved the highest performance across all classification tasks. In contrast, no single LLM dominated every task: GPT-OSS-20B obtained the best results for anxiety detection, GPT-5.2 for eating disorders, and Gemini 3.1 Pro for both depression and multiclass classification. Notably, eating disorder detection yielded the highest macro F1-scores for both model families, suggesting that this task exhibits greater class separability than anxiety, depression, or the multiclass setting. This observation is consistent with the embedding visualization presented in Figure 1 and is further supported by the confusion

Table 4: Performance of the evaluated models on the multiclass classification task. Values are reported as metric [95% confidence interval]. TARa@5 is only applicable to LLMs.

Model	Macro F1	Precision	Recall	MCC	TARa@5
DistilBETO	0.818 [0.758, 0.865]	0.844 [0.778, 0.884]	0.834 [0.775, 0.875]	0.744 [0.673, 0.807]	–
RoBERTa-large-BNE	0.842 [0.778, 0.890]	0.846 [0.774, 0.895]	0.845 [0.780, 0.888]	0.786 [0.703, 0.846]	–
mDeBERTa-v3	0.790 [0.725, 0.846]	0.822 [0.762, 0.867]	0.779 [0.705, 0.838]	0.709 [0.628, 0.779]	–
Llama 3.1 8B	0.748 [0.684, 0.809]	0.752 [0.692, 0.807]	0.799 [0.739, 0.846]	0.695 [0.621, 0.767]	0.837 [0.824, 0.849]
GPT-OSS-20B	0.724 [0.656, 0.790]	0.734 [0.666, 0.801]	0.752 [0.681, 0.809]	0.655 [0.571, 0.730]	0.847 [0.835, 0.859]
DeepSeek-V3.2	0.766 [0.695, 0.826]	0.796 [0.728, 0.848]	0.760 [0.684, 0.820]	0.684 [0.594, 0.755]	0.883 [0.872, 0.894]
GPT-5.2	0.790 [0.728, 0.846]	0.809 [0.743, 0.861]	0.820 [0.776, 0.863]	0.728 [0.651, 0.791]	0.868 [0.856, 0.878]
GPT-5 Mini	0.783 [0.719, 0.844]	0.793 [0.726, 0.851]	0.824 [0.772, 0.864]	0.735 [0.653, 0.797]	0.863 [0.851, 0.874]
Claude Sonnet 4.6	0.782 [0.714, 0.838]	0.797 [0.733, 0.851]	0.830 [0.777, 0.868]	0.745 [0.665, 0.802]	0.878 [0.867, 0.889]
Claude Haiku 4.5	0.820 [0.759, 0.871]	0.823 [0.759, 0.875]	0.844 [0.788, 0.885]	0.762 [0.685, 0.825]	0.889 [0.878, 0.899]
Gemini 3.1 Pro	0.848 [0.796, 0.901]	0.838 [0.780, 0.893]	0.862 [0.809, 0.906]	0.799 [0.733, 0.864]	0.885 [0.874, 0.896]
Gemini 3 Flash	0.849 [0.796, 0.899]	0.843 [0.785, 0.896]	0.868 [0.817, 0.910]	0.809 [0.740, 0.865]	0.872 [0.860, 0.882]

matrix analysis of the multiclass task presented later in this section.

3.2. Multiclass Results

3.2.1. Classification Metrics and Prediction Stability

Table 4 reports the performance of all evaluated models on the multiclass classification task. For each model, the table includes macro F1-score, precision, recall, MCC, and their corresponding 95% confidence intervals. For the LLMs, prediction stability is additionally reported through the TARa@5 metric together with its 95% Bayesian credible interval.

Overall, the results indicate that both BERT-based encoders and recent LLMs achieve strong performance in the multiclass setting, with relatively small differences between the best-performing models. The highest macro F1-score was obtained by Gemini 3 Flash (0.849), closely followed by Gemini 3.1 Pro (0.848) and RoBERTa-large-BNE (0.842). The substantial overlap between their confidence intervals already suggests that these differences are likely to be small, a hypothesis that is formally examined in the paired statistical comparison presented later in this section.

Among the encoder-based models, RoBERTa-large-BNE consistently outperformed DistilBETO and mDeBERTa-v3 across all reported metrics, confirming the effectiveness of large Spanish-specific pretrained encoders for this task. Remarkably, its performance remained highly competitive with the best proprietary LLMs despite requiring substantially fewer computational resources during inference.

The results also reveal considerable variability across the evaluated LLMs. While the Gemini models achieved the highest predictive performance, Claude Haiku 4.5 obtained results comparable to substantially larger proprietary

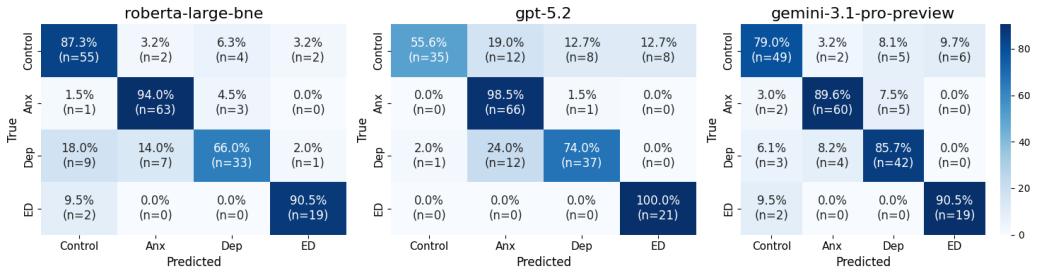


Figure 3: Representative confusion matrices for the multiclass classification task. Rows correspond to the true class and columns to the predicted class. Values indicate row-normalized recall percentages, with the absolute number of instances shown in parentheses.

models, outperforming both GPT-5.2 and Claude Sonnet 4.6 in terms of macro F1-score. In contrast, the open-source LLMs generally achieved lower performance, although DeepSeek-V3.2 consistently outperformed both GPT-OSS-20B and Llama 3.1 8B.

Regarding prediction stability, all evaluated LLMs exhibited high TARa@5 values, ranging from 0.837 to 0.889, indicating that repeated zero-shot predictions were highly consistent despite the inherent non-determinism of these models. DeepSeek-V3.2, Claude Haiku 4.5, and Gemini 3.1 Pro showed the highest stability, whereas Llama 3.1 8B produced the lowest agreement across repeated executions. Nevertheless, even the least stable model maintained agreement above 83%, suggesting that prediction variability had only a limited impact under the deterministic inference configuration adopted in this study.

3.2.2. Error Analysis

Although aggregate performance metrics provide an overall assessment of predictive accuracy, they do not reveal which classes are systematically confused by the evaluated models. To better understand these error patterns, Figure 3 presents the row-normalized confusion matrices for three representative models: RoBERTa-large-BNE, the best-performing BERT-based encoder; GPT-5.2, a representative frontier LLM exhibiting high predictive performance but asymmetric behaviour across classes; and Gemini 3.1 Pro, one of the best-performing models overall. The complete set of confusion matrices for all evaluated models is available in the accompanying code repository.

Across all three models, a consistent pattern emerges. Most classification

errors occur between the Control, Anxiety, and Depression classes, whereas Eating Disorders (ED) are identified considerably more accurately and are only rarely confused with the remaining categories. This behaviour is consistent with the embedding visualization presented in Figure 1, where ED forms a more clearly separated cluster while the other three classes exhibit substantial semantic overlap. These findings further suggest that the superior performance observed for ED detection may be related to its greater apparent separability in the semantic embedding space.

The three representative models nevertheless exhibit distinct error profiles. RoBERTa-large-BNE achieves high classification accuracy for the Control, Anxiety, and ED classes (87.3%, 94.0%, and 90.5%, respectively), but struggles to distinguish Depression, correctly classifying only 66.0% of the corresponding instances. Most of these errors arise from confusion with the Control (18.0%) and Anxiety (14.0%) classes.

GPT-5.2 displays a markedly different behaviour. While it almost perfectly identifies Anxiety (98.5%) and ED (100.0%), its accuracy on the Control class drops to 55.6%, with many control users being incorrectly classified as Anxiety (19.0%) or, less frequently, as Depression or ED (12.7% each). This asymmetric error distribution explains why GPT-5.2, despite its strong performance on some classes, does not achieve the highest overall macro F1-score.

In contrast, Gemini 3.1 Pro presents the most balanced confusion matrix, maintaining consistently high recall across all four classes (79.0%, 89.6%, 85.7%, and 90.5%). Unlike the other models, it does not exhibit a single clearly problematic class, resulting in the most homogeneous predictive behaviour among the representative models and explaining its superior overall performance in the multiclass setting. The few remaining misclassifications are largely confined to the Control, Anxiety, and Depression classes, while confusion involving the ED class remains minimal.

3.2.3. Paired Model Comparisons

Comparison of top-performing models. Table 5 summarizes the pairwise statistical comparisons between the top-performing models selected according to the procedure described in Section 2.4.3. For each pair, the table reports the number of test instances included in the comparison (Samples), the difference in macro F1-score together with its 95% paired bootstrap confidence interval, and the adjusted p-value obtained using the Approximate Randomization Test (ART). Comparisons involving Gemini models include slightly

Table 5: Pairwise statistical comparison of the top-performing models in the multiclass task. Asterisks (*) denote statistically significant differences after Benjamini–Hochberg correction ($p < 0.05$).

Model A	**Model B**	**Samples**	**ΔF1 (95% CI)**	**ART**
RoBERTa-large-BNE	DeepSeek-V3.2	201	0.076 [0.009, 0.147]	0.060
RoBERTa-large-BNE	Gemini 3 Flash	199	-0.009 [-0.068, 0.051]	0.917
RoBERTa-large-BNE	Gemini 3.1 Pro	199	-0.007 [-0.066, 0.052]	0.917
DeepSeek-V3.2	Gemini 3 Flash	199	-0.086 [-0.146, -0.028]	**0.017***
DeepSeek-V3.2	Gemini 3.1 Pro	199	-0.084 [-0.146, -0.027]	**0.017***
Gemini 3 Flash	Gemini 3.1 Pro	199	0.002 [-0.030, 0.033]	0.917

fewer samples because a small number of test instances were automatically blocked by the Gemini API under its `PROHIBITED_CONTENT` safety policy, primarily for texts containing references to self-harm or suicide.

The statistical analysis reveals that only two pairwise comparisons reached statistical significance after controlling the false discovery rate. In both cases, the Gemini models significantly outperformed DeepSeek-V3.2, with macro F1 improvements of approximately 0.08 points. By contrast, no statistically significant differences were found between the two Gemini models, nor between either Gemini model and RoBERTa-large-BNE, despite the slightly higher macro F1-scores achieved by the former.

These results highlight that small differences in aggregate performance metrics should be interpreted with caution. Although Gemini 3 Flash obtained the highest macro F1-score in the multiclass task, its performance cannot be considered statistically superior to that of RoBERTa-large-BNE or Gemini 3.1 Pro. Conversely, the statistically significant gap observed with DeepSeek-V3.2 indicates a more substantial difference in predictive performance.

Intra-family comparison of proprietary LLMs. Table 6 presents the statistical comparison between flagship proprietary LLMs and their lightweight counterparts.

No statistically significant differences were observed between any flagship proprietary model and its lightweight counterpart. In particular, GPT-5.2 and GPT-5 Mini produced virtually identical results, while Gemini 3.1 Pro and Gemini 3 Flash achieved statistically indistinguishable performance despite their different computational capabilities.

An especially noteworthy finding concerns the Claude family. Although

Table 6: Intra-family comparison of proprietary LLMs in the multiclass task.

Model A	**Model B**	**Samples**	**ΔF1 (95% CI)**	**ART**
GPT-5.2	GPT-5 Mini	201	0.007 [-0.042, 0.057]	0.917
Claude Sonnet 4.6	Claude Haiku 4.5	201	-0.038 [-0.091, 0.012]	0.462
Gemini 3.1 Pro	Gemini 3 Flash	199	-0.002 [-0.033, 0.030]	0.917

Claude Haiku 4.5 obtained a higher macro F1-score than Claude Sonnet 4.6 (0.820 vs. 0.782), this apparent advantage was not statistically significant. Together, these results suggest that, for the multiclass task considered in this study, the additional computational resources required by flagship proprietary models do not necessarily translate into measurable improvements in predictive performance.

3.2.4. Cost–Performance Analysis

Figure 4 illustrates the relationship between predictive performance and computational cost for the multiclass task. Cost estimation follows the methodology described in Section 2.4.4. The Pareto frontier identifies the set of models for which no alternative simultaneously achieves both lower cost and higher predictive performance.

However, the Pareto frontier is constructed from point estimates of macro F1-score and does not account for the uncertainty associated with model performance. Consequently, differences between models appearing on the frontier should be interpreted together with the confidence intervals and the paired statistical comparisons presented in the previous subsection. As shown by the ART analysis, some models occupying different positions on the Pareto frontier do not exhibit statistically significant differences in predictive performance.

Several interesting patterns emerge from this analysis. First, despite being based on supervised fine-tuning, RoBERTa-large-BNE remains on the Pareto frontier, offering a highly competitive compromise between predictive performance and computational cost. Although Gemini 3 Flash achieves a marginally higher macro F1-score, the improvement is relatively small compared with the additional inference cost associated with API-based deployment.

Among the evaluated LLMs, Gemini 3 Flash provides the most favourable cost-performance trade-off, combining the highest predictive performance

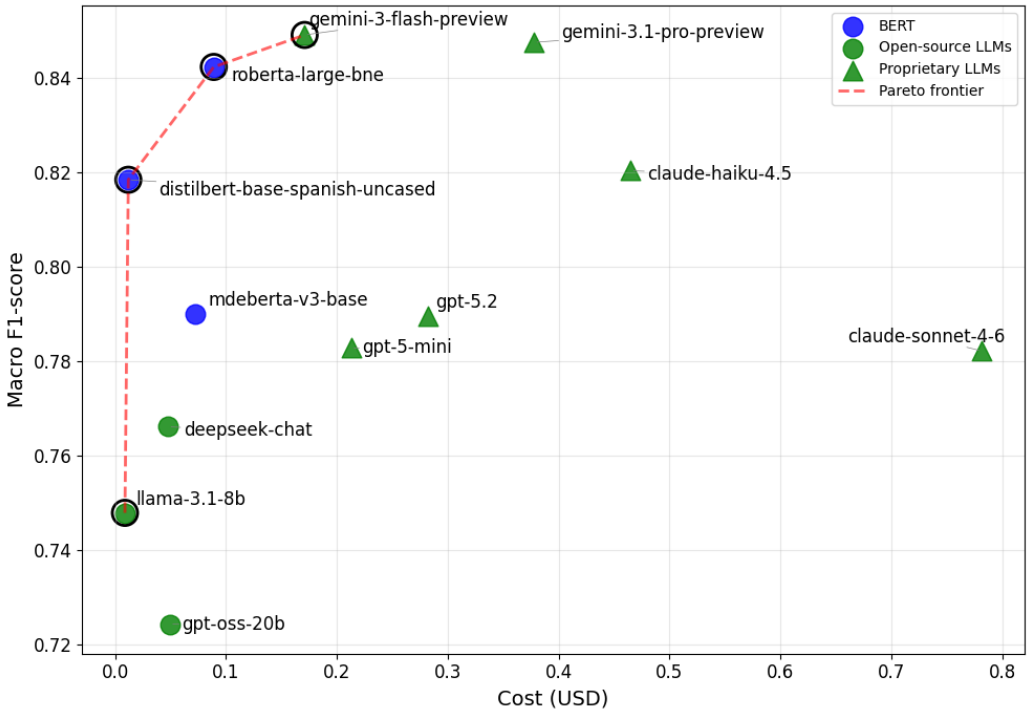


Figure 4: Cost-performance trade-off and Pareto frontier for the multiclass classification task.

with a substantially lower cost than Gemini 3.1 Pro, GPT-5.2, and Claude Sonnet 4.6. In contrast, these larger proprietary models are Pareto-dominated, indicating that their higher computational cost is not compensated by proportional improvements in classification performance.

The analysis also highlights that not all open-source LLMs constitute cost-effective alternatives. Although Llama 3.1 8B appears on the Pareto frontier because of its very low inference cost, GPT-OSS-20B and DeepSeek-V3.2 (deepseek-chat) are both dominated by RoBERTa-large-BNE, which simultaneously achieves higher predictive performance at a comparable or lower estimated cost.

Finally, DistilBETO (distilbert-base-spanish-uncased) represents the lowest-cost solution among all evaluated models and therefore occupies the leftmost point of the Pareto frontier. While its predictive performance is below that of RoBERTa-large-BNE and Gemini 3 Flash, it remains an attractive option for resource-constrained scenarios. Overall, these results indicate that the most

expensive models do not necessarily provide the best practical value, and that carefully selected encoder-based models or lightweight frontier LLMs may offer a substantially better balance between predictive performance and computational cost.

4. Discussion

Overall, recent LLMs achieved the highest predictive performance across all classification tasks, although their advantage over the best BERT-based encoder was generally modest and varied across tasks. RoBERTa-large-BNE remained particularly competitive in depression, eating disorder, and multiclass classification, demonstrating that well-established encoder-based models continue to represent strong alternatives to frontier LLMs for supervised mental health classification.

The error analysis revealed a consistent pattern across model families. Most misclassifications occurred between the Control, Anxiety, and Depression classes, whereas Eating Disorders were identified with substantially higher accuracy. This behaviour closely matched the embedding visualization presented in Figure 1, where ED formed a more clearly separated cluster while the remaining classes exhibited greater semantic overlap. These findings suggest that the main challenge in multiclass mental health detection is distinguishing conditions with similar linguistic manifestations, particularly anxiety and depression.

The statistical comparisons further showed that most differences among the strongest models were not statistically significant after controlling for multiple testing. In particular, neither Gemini model significantly outperformed RoBERTa-large-BNE, despite their marginally higher macro F1-scores. Thus, model rankings based solely on point estimates may overstate differences in predictive capability.

4.1. Theoretical and Practical Implications

From a broader research perspective, these findings support a multidimensional approach to language model evaluation. Predictive performance alone provides an incomplete picture of model suitability, particularly when differences between top-performing models are small. Complementary analyses of statistical significance, prediction stability, error patterns, and computational cost provide a more comprehensive basis for model comparison and selection.

Table 7: Comparison with previous studies evaluated on the MentalRiskES benchmark.

Study	Anxiety	Depression	EDs	Multiclass
Mármol-Romero et al. (2024)	0.710	0.700	0.810	–
Pan et al. (2023)	–	0.737	0.918	–
This work	**0.900**	**0.842**	**0.979**	**0.849**

From a practical perspective, the cost-performance analysis revealed that higher computational cost does not necessarily translate into better predictive performance. Lightweight proprietary models consistently achieved results comparable to those of their flagship counterparts while requiring substantially lower inference costs. This trend was particularly evident within the Gemini, GPT, and Claude families, where none of the larger models demonstrated statistically significant improvements over their lightweight variants. Consequently, for many real-world applications, smaller proprietary LLMs may offer more efficient deployment options when API-based inference is preferred.

The analysis also highlights the continued practical relevance of encoder-based models. Despite requiring an initial fine-tuning stage, RoBERTa-large-BNE remained on the Pareto frontier and achieved performance statistically indistinguishable from the best proprietary LLMs in the multiclass task. Since the computational cost of encoder-based models is largely incurred during training, while inference remains comparatively inexpensive, these models become increasingly attractive as the number of processed documents grows. Consequently, the choice between encoder-based models and LLMs should not be based solely on predictive performance, but also on the intended deployment scenario, the expected inference volume, and the available computational budget.

4.2. Comparison with Previous Work

Table 7 compares the best results obtained in this study with those reported in the two previous works that evaluated models on the MentalRiskES benchmark. Since all three studies employ the same dataset, this comparison provides a direct assessment of the progress achieved in Spanish mental health disorder detection.

The best-performing models in this study outperformed all directly comparable results previously reported on MentalRiskES. Compared with Mármol-

Romero et al., macro F1 increased from 0.710 to 0.900 for anxiety, from 0.700 to 0.842 for depression, and from 0.810 to 0.979 for eating disorder detection. Improvements were also observed over the transformer-based results reported by Pan et al. for depression (0.737 vs. 0.842) and eating disorders (0.918 vs. 0.979). These gains likely reflect advances in pretrained language models and the inclusion of recent proprietary and open-source LLMs unavailable in earlier studies.

Beyond predictive performance, this work extends previous research by incorporating prediction stability analysis, statistical significance testing, detailed error analysis, and a systematic assessment of the relationship between predictive performance and computational cost. These complementary analyses provide a more comprehensive understanding of the strengths and limitations of contemporary language models for mental health disorder detection.

Direct comparisons with studies using different datasets should be interpreted with caution because of differences in language, annotation protocols, class distributions, and task definitions. Nevertheless, our findings are broadly consistent with recent evidence that modern LLMs achieve competitive or superior performance to traditional encoder-based models for mental health text classification.

4.3. Limitations and Future Work

The principal limitation of this study is the size of the MentalRiskES dataset. Although it represents one of the few available and carefully annotated benchmarks for Spanish mental health disorder detection, the relatively small number of users limits both the generalizability of the reported results and the statistical power of the pairwise comparisons. Larger datasets would not only enable more robust statistical analyses but also provide additional training data for encoder-based models, which are likely to benefit more substantially from supervised fine-tuning than foundation LLMs. Furthermore, extending the evaluation to clinically validated datasets or annotations provided directly by mental health professionals would strengthen the clinical relevance of future studies.

Several methodological extensions also warrant further investigation. First, this work focused exclusively on zero-shot prompting for LLMs. Exploring alternative inference strategies, such as few-shot prompting or more advanced prompt engineering techniques, may further improve predictive performance without requiring additional model training. Second, retrieval-augmented generation (RAG) represents a promising direction, as incorporating external

clinical knowledge could provide richer contextual information and improve both prediction accuracy and interpretability.

Finally, the rapid evolution of language models suggests that the conclusions drawn in this work should be continuously reassessed as new generations of proprietary and open-source LLMs become available. Future studies should also evaluate larger multilingual datasets and investigate additional aspects such as model explainability, calibration, and fairness, which are essential for the responsible deployment of AI systems in mental health applications.

4.4. Ethical Considerations

The models evaluated in this study should be understood as detecting linguistic indicators of mental health risk rather than providing clinical diagnoses, since the underlying annotations were produced by non-professional annotators rather than clinicians. Their outputs should therefore never substitute for assessment by qualified mental health professionals. The error analysis further revealed asymmetric misclassification patterns with direct ethical implications: models such as GPT-5.2 frequently misclassified Control users as Anxiety or Depression, illustrating the risk of falsely identifying mental health concerns, while false negatives may overlook users who genuinely need support. Both risks underscore the need for human oversight in any real-world deployment.

Given the sensitivity of this domain, future research involving real user data should undergo appropriate ethical review and informed consent procedures, with particular attention to data privacy and access-control safeguards, especially when using third-party proprietary APIs that process user-generated text externally.

5. Conclusion

This work presented a comprehensive evaluation of BERT-based encoders, open-source LLMs, and proprietary frontier LLMs for automatic detection of anxiety, depression, and eating disorders in Spanish using the MentalRiskES benchmark. By combining predictive performance, prediction stability, statistical significance testing, error analysis, and cost-performance assessment within a unified evaluation framework, the study provides a more comprehensive benchmark than previous work on Spanish mental health text classification.

The proposed evaluation methodology is independent of the specific models considered and can be readily extended to future generations of language models and new mental health datasets. We hope that this benchmark and evaluation framework contribute to more rigorous and practically relevant assessments of language models for mental health applications.

CRediT authorship contribution statement

Andrés Corbacho-Sánchez: Conceptualization, Methodology, Software, Investigation, Data curation, Formal analysis, Validation, Visualization, Writing – original draft, Writing – review & editing. **Adrián Domínguez-Díaz:** Conceptualization, Methodology, Supervision, Validation, Writing – review & editing. **Luis de Marcos:** Conceptualization, Methodology, Supervision, Validation.

Data availability

The MentalRiskES dataset used in this study is not publicly available and can be obtained from its original authors subject to their data access policy.

Code availability

The source code, prompts, evaluation scripts, and experimental framework used in this study are publicly available at: `https://github.com/AndresCorSan/mental-health-bert-llm`.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work, the authors used ChatGPT (OpenAI) in order to improve the clarity, organization, and academic writing of the manuscript. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

Funding

This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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